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main.py

```python
def find_min_max(arr, low, high):
    if low == high:
        return arr[low], arr[low]

    mid = (low + high) // 2

    min1, max1 = find_min_max(arr, low, mid)
    min2, max2 = find_min_max(arr, mid + 1, high)

    return min(min1, min2), max(max1, max2)

arr = [120, 150, 98, 175, 140, 165, 110, 190, 130, 145]

minimum, maximum = find_min_max(arr, 0, len(arr) - 1)

print("Minimum Height =", minimum)
print("Maximum Height =", maximum)
```

input

```
Minimum Height = 98
Maximum Height = 190


...Program finished with exit code 0
Press ENTER to exit console.
```

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main.py

```python
def find_min_max(arr, low, high):
    if low == high:
        return arr[low], arr[low]

    mid = (low + high) // 2

    min1, max1 = find_min_max(arr, low, mid)
    min2, max2 = find_min_max(arr, mid + 1, high)

    return min(min1, min2), max(max1, max2)

arr = [14, 12, 18, 11, 15, 13, 17, 16, 10]

minimum, maximum = find_min_max(arr, 0, len(arr) - 1)

print("Fastest Time =", minimum)
print("Slowest Time =", maximum)
```

input

```
Fastest Time = 10
Slowest Time = 18


...Program finished with exit code 0
Press ENTER to exit console.
```

main.py

```python
def find_min_max(arr, low, high):
    if low == high:
        return arr[low], arr[low]

    mid = (low + high) // 2

    min1, max1 = find_min_max(arr, low, mid)
    min2, max2 = find_min_max(arr, mid + 1, high)

    return min(min1, min2), max(max1, max2)

arr = [450, 520, 480, 610, 430, 590]

minimum, maximum = find_min_max(arr, 0, len(arr) - 1)

print("Minimum Stock Price =", minimum)
print("Maximum Stock Price =", maximum)
```

input

```
Minimum Stock Price = 430
Maximum Stock Price = 610


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Press ENTER to exit console.
```

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main.py

```python
def find_min_max(arr, low, high):
    if low == high:
        return arr[low], arr[low]

    mid = (low + high) // 2

    min1, max1 = find_min_max(arr, low, mid)
    min2, max2 = find_min_max(arr, mid + 1, high)

    return min(min1, min2), max(max1, max2)

arr = [78, 92, 65, 88, 95, 72, 81, 69]

minimum, maximum = find_min_max(arr, 0, len(arr) - 1)

print("Minimum Mark =", minimum)
print("Maximum Mark =", maximum)
```

input

```
Minimum Temperature = 25
Maximum Temperature = 37


...Program finished with exit code 0
Press ENTER to exit console.
```

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main.py

```python
def find_min_max(arr, low, high):
    if low == high:
        return arr[low], arr[low]

    mid = (low + high) // 2

    min1, max1 = find_min_max(arr, low, mid)
    min2, max2 = find_min_max(arr, mid + 1, high)

    return min(min1, min2), max(max1, max2)

arr = [32, 28, 35, 25, 31, 29, 37]

minimum, maximum = find_min_max(arr, 0, len(arr) - 1)

print("Minimum Temperature =", minimum)
print("Maximum Temperature =", maximum)
```

input

```
Minimum Temperature = 25
Maximum Temperature = 37


...Program finished with exit code 0
Press ENTER to exit console.
```

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main.py

```python
arr = [1, 2, 2, 2, 3, 4, 5, 6]
key = 2

low = 0
high = len(arr) - 1
result = -1

while low <= high:
    mid = (low + high) // 2

    if arr[mid] == key:
        result = mid
        high = mid - 1
    elif key < arr[mid]:
        high = mid - 1
    else:
        low = mid + 1

if result != -1:
    print("First occurrence at index", result)
else:
    print("Element not found")
```

input

```
First occurrence at index 1


...Program finished with exit code 0
Press ENTER to exit console.
```

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main.py

```python
arr = [5, 10, 15, 20, 25, 30, 35]
key = 25

low = 0
high = len(arr) - 1
count = 0
found = False

while low <= high:
    count += 1
    mid = (low + high) // 2

    if arr[mid] == key:
        print("Element found")
        found = True
        break
    elif key < arr[mid]:
        high = mid - 1
    else:
        low = mid + 1

if not found:
    print("Element not found")

print("Iterations =", count)
```

input

```
Element found
Iterations = 3


...Program finished with exit code 0
Press ENTER to exit console.
```

main.py

```python
arr = [2, 4, 6, 8, 10, 12]
key = 8

low = 0
high = len(arr) - 1
index = -1

while low <= high:
    mid = (low + high) // 2

    if arr[mid] == key:
        index = mid
        break
    elif key < arr[mid]:
        high = mid - 1
    else:
        low = mid + 1

print("Index =", index)
```

```
Index = 3


...Program finished with exit code 0
Press ENTER to exit console.
```

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main.py

```python
arr = [10, 20, 30, 40, 50]
key = 30

low = 0
high = len(arr) - 1
found = False

while low <= high:
    mid = (low + high) // 2

    if arr[mid] == key:
        print("Element found at position", mid + 1)
        found = True
        break
    elif key < arr[mid]:
        high = mid - 1
    else:
        low = mid + 1

if not found:
    print("Element not found")
```

input

```
Element found at position 3


...Program finished with exit code 0
Press ENTER to exit console.
```

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